

Predation rates based on dummy caterpillars

Shirin Shoghli, Shahrzad Torabi, Sanaz Bayat, Maryam Gheibi, Mozhdeh Ramezani Dastjerdi

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5 Week 07 (December 22 - January 11)

Our 3 datasets:

<https://www.bexis.uni-jena.de/ddm/data/Showdata/31113>

<https://www.bexis.uni-jena.de/ddm/data/Showdata/24246>

<https://www.bexis.uni-jena.de/ddm/data/Showdata/25807>

Interoperable

Formal Knowledge Representation:

- The dataset can be applied and utilized in related fields as well.
- The language used in the dataset is designed to be easily understood by the community.
- The BExIS dataset offers metadata representation in formal XML and HTML formats.
- It provides a comprehensive data structure with detailed data types and attribute categories.
- The dataset is available in various file formats, such as csv, tsv, xlsx, and txt.

Vocabulary Adherence to FAIR Principles:

- There are no glossaries for the presented datasets that explains the professional terms.
- However, metadata in the datasets contains title, abstract, etc. which match the FAIR format.

Title	Predation marks on dummy caterpillars measured in grassland experiments (RPs and UPs, including corresponding EPs) in 2021	i
Abstract	In May 2021, we investigated predation on dummy caterpillars over 48 h in all three Exploratory regions. We investigated (i) whether plot-level process rates and their spatial variation differed among treatments of the grasslands experiments (RPs and UPs), (ii) whether process rates were correlated among each other across sites and (iii) whether the rates were affected by climate and land-use intensity at short and medium-term temporal scales. The data contain the presence/absence of attack marks separated by arthropods, small mammals, birds and snails. As potential covariates, vegetation cover, vegetation height and canopy cover are given.	i
Project name	Arthropods (core)	i
Metadata creation date	10/26/2021 2:38:59 PM	i
Metadata last modification date	11/16/2021 11:00:16 AM	i
Data last modification date	10/26/2021 2:45:40 PM	i
Id	31113	i
Version	4	i
Metadata schema	Biodiversity Exploratories Metadata Schema 3.0	i
License	CC BY 4.0	i
Funder name	German Research Foundation	i
Funder identifier	https://ror.org/018mejw64	i
Metadata creator	Michael Staab	i
Data creator	Michael Staab	i
Data collector	Michael Staab	i

Qualified References to Other (Meta)Data:

- There are no references to other metadata or any links to connect our dataset to a different dataset/metadata which point to a relationship between them.

Challenges and Improvements

As we mentioned above:

1. There are no documents attached to datasets that explain the professional words or terms related to the dataset.
2. The datasets do not include any references or links that connect them to other datasets or metadata, thereby indicating a relationship between them is absent.

Proposed Enhancement:

1. Creating a glossary according to the dataset and special words or statements that may be hard to comprehend by non-experts.
2. The above solution can also be used for references.

Reusable

Attributes and Accuracy:

Take a look at the dataset documentation available on the BEXIS data portal.

- The documentation includes a dedicated tab for the Data Structure.
- The variable names mentioned in the dataset accurately reflect the values stored in the data.
- Upon reviewing the values, it is evident that they are saved by the specified data structures.
- The saved attributes specifically pertain to the subject of interest, which in this case is dummy caterpillars.



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[SEIS - Show Dataset: Predation marks on dummy caterpillars measured in grassland experiments \(RPs and UPs, including corresponding EPs\) in 2021](#)

Predation marks on dummy caterpillars measured in grassland experiments (RPs and UPs, including corresponding EPs) in 2021

Dataset id: 31113 Version: 4 Metadata edited (16.11.2021)

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[Metadata](#) | [Primary Data](#) | [Data Structure](#) | [Links](#) | [Attachments](#)

[Download Excel](#) | [Download Text](#) | ☐ Download data with units ☒ Fit table to screen.

Dummy_ID	Region	PlotID	Basis_PlotID	Treatment	Subplot	Dummy	Arthropod...	Bird_mark	Mammal...	Gastropod...	Predation...	Dummy_m...	Start_date	End_date	Max_veg...	Average_v...	Cover_veg	Cover_can...
AE001-EP_E1	A	AE001	AE01	EP	E	1	0	0	0	0	0	0	2021-05-24	2021-05-26	50	30	95	0
AE001-EP_E2	A	AE001	AE01	EP	E	2	0	0	0	0	0	0	2021-05-24	2021-05-26	50	30	95	0
AE001-EP_E3	A	AE001	AE01	EP	E	3	0	1	0	0	1	0	2021-05-24	2021-05-26	50	30	95	0
AE001-EP_E4	A	AE001	AE01	EP	E	4	0	1	0	0	1	0	2021-05-24	2021-05-26	50	30	95	0
AE001-EP_E5	A	AE001	AE01	EP	E	5	1	0	0	0	1	0	2021-05-24	2021-05-26	50	30	95	0
AE001-EP_E6	A	AE001	AE01	EP	E	6	1	0	0	0	1	0	2021-05-24	2021-05-26	50	30	95	0
AE001-EP_E7	A	AE001	AE01	EP	E	7	0	0	0	0	0	0	2021-05-24	2021-05-26	50	30	95	0
AE001-EP_E8	A	AE001	AE01	EP	E	8	0	0	0	0	0	0	2021-05-24	2021-05-26	50	30	95	0
AE001-EP_E9	A	AE001	AE01	EP	E	9	0	0	0	0	0	0	2021-05-24	2021-05-26	50	30	95	0
AE001-EP_E10	A	AE001	AE01	EP	E	10	0	0	0	0	0	0	2021-05-24	2021-05-26	50	30	95	0

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SEIS - Show Dataset: Predation marks on dummy caterpillars measured in grassland experiments (RPs and UPs, including corresponding EPs) in 2021

Dataset id: 31113
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Download Dataset
Download Citation

Metadata
Primary Data
Data Structure
Links
Attachments

30904 data structure V2 (1368)

Data structure for structured data

Variable Name	Description	Unit	Data Type	Category	Optional	Variable Id
Dummy_ID	Combination of Plot, subplot and dummy number	none	string	id string	☒	23358
Region	Name of the Exploratory	none	string	id string	☒	23359
PlotID	Experimental plot with leading 0	none	string	id string	☒	23360
Bevis_PlotID	Experimental plot	none	string	plot id	☒	23361
Treatment	Treatment of grassland experiment (EP): Experimental ...	none	string	id string	☐	23362
Subplot	Experimental subplot, in all experimental plots, there we...	none	string	id string	☒	23363
Dummy	Dummy number per subplot	none	integer	quantity integer	☒	23377
Arthropod_mark	1 = mark by arthropods present, 0 = no mark by arthrop...	none	string	bool character	☒	23365
Bird_mark	1 = mark by birds present, 0 = no mark by birds.	none	string	bool character	☒	23366
Mammal_mark	1 = mark by mammals present, 0 = no mark by mammals.	none	string	bool character	☒	23367
Gastropod_mark	1 = mark by gastropods present, 0 = no mark by gastro...	none	string	bool character	☒	23368
Predation_total	1 = mark of any type present, 0 = no mark.	none	string	bool character	☒	23369
Dummy_missing	1 = dummy could not be retrieved from the field	none	string	bool character	☒	23370
Start_date	Date on which the dummy was exposed	none	date-iso	date ISO	☒	23371
End_date	Date on which the dummy was retrieved	none	date-iso	date ISO	☒	23372
Max_veg_height	Maximum height of the vegetation in the 1 m² subplot o...	cm	double	length	☒	23373
Average_veg_height	Average vegetation height in the 1 m² subplot on which ...	cm	double	length	☒	23374
Cover_veg	Cover of vegetation in the 1 m² subplot on which proces...	%	double	parts-per expression	☐	23375
Cover_canopy	Cover of the canopy above the 1 m² subplot on which pr...	%	double	parts-per expression	☐	23376

Data Usage License:

Verifying whether the dataset is made available with a transparent and easily accessible data usage license.

- It has been openly shared under the CC BY 4.0 license.
- The license reference is also provided, offering clear guidelines on sharing, adapting, and the legal aspects, explicitly explained and documented.



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Project name	Arthropods (core)	
Metadata creation date	10/26/2021 2:38:59 PM	
Metadata last modification date	11/16/2021 11:00:16 AM	
Data last modification date	10/26/2021 2:45:40 PM	
Id	31113	
Version	4	
Metadata schema	Biodiversity Exploratories Metadata Schema 3.0	
License short name, e.g. CC BY 4.0	CC BY 4.0 (Creative Commons Attribution 4.0 International).	
License	CC BY 4.0	

Provenance:

The metadata Abstract of the dataset provides additional information about the dataset's provenance. Provenance refers to the origin or source of the dataset, including details about its creation, collection, and any transformations or modifications it has undergone. Understanding the dataset's provenance is important for assessing its reliability, credibility, and suitability for specific purposes.

In addition to the provenance details, the dataset also includes contact information for the metadata moderator. The metadata moderator is responsible for overseeing and managing the dataset's metadata, ensuring its accuracy, completeness, and adherence to standards. They serve as a point of contact for any inquiries, feedback, or clarifications related to the dataset.

Having access to the metadata moderator's contact information can be valuable for users who may have questions about the dataset, need assistance in interpreting the metadata, or wish to provide feedback or report any issues. It facilitates communication and collaboration between dataset users and those responsible for maintaining the dataset's metadata quality and integrity.

✖ Contacts

Metadata creator

Michael Staab

i

Data creator

Michael Staab

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Data collector

Michael Staab

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Project leader

Nico Blüthgen

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✖ Contact Person

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Title

Predation marks on dummy caterpillars measured in grassland experiments (RPs and UPs, including corresponding EPs) in 2021



Abstract

In May 2021, we investigated predation on dummy caterpillars over 48 h in all three Exploratory regions. We investigated (i) whether plot-level process rates and their spatial variation differed among treatments of the grasslands experiments (RPs and UPs), (ii) whether process rates were correlated among each other across sites and (iii) whether the rates were affected by climate and land- use intensity at short and medium-term temporal scales.

The data contain the presence/absence of attack marks separated by arthropods, small mammals, birds and snails. As potential covariates, vegetation cover, vegetation height and canopy cover are given.



Project name

Arthropods (core)



Community Standards:

The dataset can be accessed through the BEXIS data portal.

- The dataset is associated with a publication.
- This association indicates the dataset's compliance with the accepted standards within the community.

* datasets only	
Taxonomic Term	
☐ Animals - Invertebrates (1)	
☐ Animals - Vertebrates (1)	
Project	
☐ Anthropods (core) (1)	
Consortia *	

All	Predation marks on dummy caterpillars measured	Search
All : Predation marks on dummy caterpillars measured in grassland experiments (core) and UPIs, including corresponding EPs) in 2021		
ID	Title	Project
31113	Predation marks on dummy caterpillars measured in grassland experiments (...)	Anthropods (core)
Start Date		End Date
2021-05-11		2021-05-28

Challenges and Improvement

There were no challenges encountered in this part

Proposed Enhancement:

It is possible to generate metadata that provides information about the provenance of the data, including details about the data collection methods, instruments utilized, and other relevant information.

We also used F-UJI which is an open-source tool for evaluating the FAIRness of the dataset. The result can be seen in the following picture.

Assessment Results:

Evaluated Resource:

FAIR level: ②		initial
Resource PID/URL:	https://www.bexis.uni-jena.de/ddm/data/Showdata/31113	
DataCite support:	enabled	
Metric Version:	metrics_v0.5	
Metric Specification:	https://doi.org/10.5281/zenodo.6461229	
Software version:	3.1.0	
Download assessment results:	(JSON)	
Save and share assessment results:		

Summary:

		Score earned:	Fair level:
	Findable:	1 of 7	initial
	Accessible:	0 of 3	incomplete
	Interoperable:	0 of 4	incomplete
	Reusable:	0 of 10	incomplete

Contributions

Shirin Shoghli

Checking the dataset for interoperability and reusability properties such as assessing adherence to FAIR principles, evaluating vocabularies and references, verifying attribute accuracy, ensuring clear data usage licenses, and documenting provenance.

Shahrzad Torabi

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Sanaz Bayat

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Maryam Gheibi

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